Solving time dependent Maxwell's Equations and 1D and 2D FDTD Simulation and Analysis of Electromagnetic Wave Propagation in Nonlinear Photonic Crystals

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1 Introduction

Title MW/M31 Galaxy Major Merger Remnant: Stellar remnant and formation of elliptical galaxies

Specific Research Questions - Kinematics of the Merged Bulges and Disks: evolution of the rotation curve ("ob- served" and mass-derived), dispersion profile, angular momentum. How well is the remnant described as a classical elliptical galaxy based on kinematics? Is there any rotation?

Proposed Topic - Analysing the kinematics of the MW/M31 Galaxy Major Merger Remnant.

The study of major galaxy mergers and their kinematic impact on the remnant structure is a very important part of the galaxy evolution studies. By analyzing the kinematics of merger remnants, we can trace the evolutionary pathways that lead to different structural and dynamical outcomes. The formation of ellipticals due to Major Mergers is something which can be found by analyzing the kinematics of the remnant structure. Some of the results by analysing the kinematics are as follows. Compared to the remnants of dissipationless disc galaxy mergers, the remnants created by the collision of gas-rich disc galaxies are rounder, smaller, and have bigger rotation speeds and higher centre velocity dispersions. Dissipational merger may also provide a natural mechanism to produce kinematic subsystems in elliptical galaxies (Barnes *et al.*, 1991)

One important thing that we know how gas-rich mergers can trigger intense starbursts, followed by quenching due to AGN and supernova feedback. The figure shows the Milky Way–Andromeda merger, highlighting our understanding of how an intragroup medium (IGM) accelerates mergers by enhancing dynamical friction. This process influences the final remnant's kinematics, likely resulting in an elliptical galaxy. (Cox and Loeb, 2008)

Some of the open questions include how surface brightness profiles, metallicities and abundance ratios, and kinematic subsystems of our merger remnants of very distant galaxies and how it can used to further enhance our explanation of galaxy evolution.(Cox *et al.*, 2006)

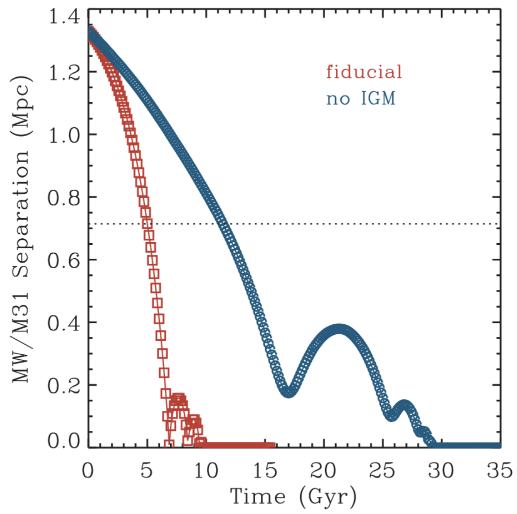

Figure 8. The separation between Andromeda and the MW during the course of their merger. Our fiducial model in red is compared to a case with no intragroup medium (dark matter + gas) in blue. Dynamical friction on the diffuse medium shortens the MW/M31 coalescence time by a factor of a few. The present time is 5 Gyr after the start of the simulation.

Figure 1: The separation between the centres of Andromeda and the MW during the course of their merger.

2 Method

- To study the kinematics of the merger remnant, I will first identify simulation snapshots where the MW and M31 have fully merged and the system is dynamically relaxed.
- These snapshots will by the seperation plots in Homework 6. Using the formula (SN * 10)/0.7, I found out the snapshots numbers to be 280, 420 and 600. I will use high res as it would give me much better velocity dispersions as a function of r.
- Extract stellar disk and bulge particles from both MW and M31.
- Concatenate them.
- Compute the center of mass position (**Reference: HW 2**) (R_{COM}) and velocity (V_{COM}) using:

$$\mathbf{R}_{\text{COM}} = \frac{\sum m_i \mathbf{r}_i}{\sum m_i}, \quad \mathbf{V}_{\text{COM}} = \frac{\sum m_i \mathbf{v}_i}{\sum m_i} \tag{1}$$

- Calculate positions and velocities to be in the newly computed COM reference frame:

$$\mathbf{r}_i' = \mathbf{r}_i - \mathbf{R}_{\text{COM}}, \quad \mathbf{v}_i' = \mathbf{v}_i - \mathbf{V}_{\text{COM}} \tag{2}$$

- Then I would orient the galaxy edge on by orienting the angular momentum along the z axis and create a phase diagram (V_θ vs. radius), with color coding to understand the direction of rotation and if at all rotation is happening. (Important thing is to extend the area to around 10-15 kpc to observe any true rotational effects)(**Reference: Lab 7**)
- Also, will calculate the observed and mass-derived rotation curves:

$$V_c(r) = \sqrt{\frac{GM(r)}{r}} \tag{3}$$

- Calculate the velocity dispersion profile using half mass radius.
- Calculate the ratio of ordered to random motion (V/σ):
- If $V/\sigma \gg 1$, the system is a disk-like fast rotator.

- If $V/\sigma \ll 1$, it is a pressure-supported elliptical galaxy.
- Also, I will the total angular momentum of the remnant:

$$\mathbf{L} = \sum m_i(\mathbf{r}_i \times \mathbf{v}_i) \tag{4}$$

- Apply the virial theorem to estimate total mass and compare it to the simulation data (**Reference: Lab 5**):
- Also I will check if the remnant follows the fundamental plane relation for elliptical galaxies and analyse it:

$$R_e \propto \sigma^2 \langle I \rangle^{-0.82} \tag{5}$$

3 Hypothesis

- The MW/M31 merger remnant might exhibit rotational support. High velocity dispersion is expected due to the violent relaxation process which would be similar to elliptical galaxies. The MW/M31 merger remnant is expected to retain some rotation, but it will likely be a slow rotator due to the low gas content of both galaxies. The remnant may form a large bulge, and stellar populations might add complexity to the rotation profile. The dark matter halos will also influence the system's dynamics, potentially dampening rotation further, leading to a slower overall spin.

References

Barnes, Joshua, Hernquist, Lars, and Schweizer, Francois (1991). "Colliding galaxies", *Scientific American*, Vol. 265, pp. 40–47. DOI: 10.1038/scientificamerican0891-40.

Cox, T. J. and Loeb, Abraham (2008). "The collision between the Milky Way and Andromeda", Vol. 386 No. 1, pp. 461–474. DOI: 10.1111/j.1365-2966.2008.13048.x. arXiv: 0705.1170 [astro-ph].

Cox, T. J. *et al.*, (2006). "The Kinematic Structure of Merger Remnants", *The Astrophysical Journal*, Vol. 650 No. 2, p. 791. DOI: 10.1086/507474. **available at**: https://dx.doi.org/10.1086/507474.